Weight-count compression by activation-weighted anchoring: an exploratory empirical study

PREPRINT REVIEW DRAFT -- no venue assumed

No established global-preservation or practical-compression claim. Original FP16 readiness remains failed. Native-tokenizer perplexities are compared within models only. Partial and cancelled attempts remain evidence, not completed cohorts.

Equations use readable ASCII notation (beta, rho, lambda; dagger denotes the pseudoinverse). The accompanying standalone manuscript.tex provides exact typeset mathematical source. This PDF is independently rendered from the same Markdown and measured figures.

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Abstract

This report examines whether matched activation-weighted low-rank approximations preserve global transformer behavior. It retains weighting/anchoring controls, full-stack and composition measurements, separate BF16/FP8 numerical controls, and the user-amended hybrid-architecture panel. Tables below are generated from retained artifacts; failed/incomplete attempts remain visible. The current scientific conclusion is inconclusive, promotion is disabled, and no universal or practical compression advantage is established. A terminal dispatcher does not imply every scientific cohort completed.

Study status and scope

Dispatcher: running architecture-panel. Report mode: interim snapshot.

The amended panel uses Gemma, Qwen3.5-0.8B and LFM2.5-1.2B-Instruct; GPT-2 is excluded. Only full-attention output projections are targeted in the hybrids. Native-tokenizer PPL is compared within each model. Validation frontier ranks and primary test ranks remain separate; no selection is made using confirmation outcomes.

Declared numerical-control omissions

The continuation formally omits unfinished dense_fp8_e4m3 and dense_fp8_e5m2 controls. All seven saved controls, including failed gates, and the cancelled attempt remain preserved. This scope closure does not claim that the original nine-control cohort completed or that native FP8 model inference was tested.

Mathematical methods and frozen constants

Let W in R^{o x i} and X in R^{i x n}. Write A=UV with rank at most r. The anchored objective is ||(W-A)X||_F^2+beta n||W-A||_F^2.

Set H=XX^T+beta nI and C=WH for exact teacher outputs. For measured outputs subtract the teacher bias first: Y_b=Y-b1^T, then C=Y_bX^T+beta nW. The extra-ridge solver adds lambda I to the geometry, with lambda=rho mean(eig(H)). For G=H+lambda I, form A_full=CG^dagger and truncate A_fullG^{1/2} to rank r; multiply back by G^{dagger/2}. Thus ridge-free full-rank anchored recovery gives W on retained spectral directions, while positive extra ridge generally shrinks it.

The implementation computes weighted covariance, eigendecomposition and weighted truncation in FP64, then returns FP32 factors for the declared storage cast. It clamps negative numerical eigenvalues to zero and retains regularized eigenvalues strictly above rcond lambda_max(G). Plain SVD uses FP32. Fixed-rank diagonal ASVD uses s_j=max(mean_t|X_{jt}|,10^{-8})^{0.5}, truncates Wdiag(s), and scales back. Factors use the recorded unbalanced convention unless a run contract explicitly declares a separate convention.

The predeclared validation anchor sweep is beta={0,0.03,0.1,0.3,1}; the extra-ridge fractions are rho={0,0.001,0.01} at beta=0.1. Validation NLL ties within 1e-6 prefer less regularization. Native evaluation uses context/stride 512/256, full shifted targets including the tail, and 16 disjoint 512-token calibration chunks for seeds 11/23/37. The primary Gemma attention-output rank is 384. The amended hybrid rank is floor((17/24)*out*in/(out+in)); frontier ranks are floor(min(out,in)/divisor) for validation-only divisors2/3/4. Exact observed run constants and factor precisions follow below; missing fields remain unrecorded.

Wide-table continuation: identical row IDs identify the same original record across every panel. IDs restart for each source table.

Panel 1 of 2; match by Row ID.

Row ID	Run	Precision	Main rank	Pseudoinverse rcond
R001	architecture-replication-20261014-082807-5c73f669	bfloat16	shape-derived by 17/24 factor/dense fraction	1e-07

Row ID	Run	Precision	Main rank	Pseudoinverse rcond
R002	baselines-20261013-175255-6a00d4e8	float16	384	1e-07
R003	composition-20261013-185512-2bffd477	float16	384	1e-07
R004	composition-20261014-032120-486d799f	float16	384	1e-07
R005	composition-20261014-054423-dbd821ff	float16	384	1e-07
R006	precision-readiness-20261014-060840-6db37daf	float16	384	1e-07
R007	regularization-20261013-171941-804f4c2e	float16	384	1e-07

Panel 2 of 2; match by Row ID.

Row ID	Run	Config beta	Config ridge fraction	Selected solver / baseline
R001	architecture-replication-20261014-082807-5c73f669	unmeasured	unmeasured	{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "solver": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}}
R002	baselines-20261013-175255-6a00d4e8	0.1	0.01	{"baseline": "SVD and fixed-rank ASVD / whitening-only components, same FP16/rank", "solver": null}
R003	composition-20261013-185512-2bffd477	0.1	0.01	{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "solver": null}
R004	composition-20261014-032120-486d799f	0.1	0.01	{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "solver": null}
R005	composition-20261014-054423-dbd821ff	0.1	0.01	{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "solver": null}
R006	precision-readiness-20261014-060840-6db37daf	0.1	0.01	{"baseline": "untouched dense BF16 teacher and copied BF16 control", "solver": null}
R007	regularization-20261013-171941-804f4c2e	0.1	0.01	{"baseline": "SVD and fixed-rank ASVD / whitening-only components, same FP16/rank", "solver": null}

Related work and scope of the planned evidence

Data-aware low-rank compression predates this study. DRONE (https://proceedings.neurips.cc/paper/2021/hash/f56de5ef149cf0aedcc8f4797031e229-Abstract.html) derives a closed-form approximation informed by layer representations in BERT [@chen2021drone]. Fisher-weighted factorization (https://arxiv.org/abs/2207.00112) uses parameter importance to address disagreement between matrix reconstruction and task accuracy [@hsu2022weighted]. These are context for the present question: low local weight error alone cannot establish preserved global computation.

ASVD (https://arxiv.org/abs/2312.05821) adapts decomposition to activation distributions and layer sensitivity [@yuan2023asvd].

SVD-LLM (https://arxiv.org/abs/2403.07378) combines truncation-aware whitening with sequential updates after compression [@wang2024svdllm]. Our fixed-rank

diagonal ASVD and whitening-only SVD-LLM isolate their weighting components under a matched factor budget. They are not full-pipeline reproductions.

SVD-LLM V2 (https://aclanthology.org/2025.naacl-long.217/) changes rank allocation and singular-value truncation [@wang2025svdllmv2], so it belongs in a later adaptive-budget study rather than a claim that fixed per-matrix ranks reproduce its complete procedure.

Global composition is already an explicit target of related work.

SAES-SVD (https://arxiv.org/abs/2602.03051) addresses accumulated and local errors [@hu2026saessvd]; LACE-SVD (https://arxiv.org/abs/2607.03057) combines loss-aware allocation with residual-stream correction [@liu2026lacesvd].

Our smaller-group and downstream-drift comparisons test the location and extent of composition damage, without claiming to resolve it. The next comparator is SAES-SVD if cumulative drift remains the decisive failure.

Quantization Error Propagation (https://arxiv.org/abs/2504.09629) demonstrates why accumulated errors also matter in layer-wise quantization [@arai2025qep]. That connection motivates measuring downstream behavior, but does not equate low-rank truncation with quantization or transfer an existing method's results.

Loss-aware global allocation provides further distinctions from this study's fixed-rank setup. Zero Sum SVD (https://arxiv.org/abs/2602.02848) uses whitened component sensitivities under a global pruning rule [@abbasi2026zerosum].

IO-SVD (https://arxiv.org/abs/2605.15626) adds an output-side predictive metric and heterogeneous rank selection [@abbasi2026iosvd]. These motivate future equal-total-budget comparisons. They do not establish novelty for an anchored input-covariance truncation or for the current extra-ridge solver.

The practical comparison distinguishes storage, accuracy, and execution cost.

GPTQ (https://arxiv.org/abs/2210.17323) uses approximate second-order information for post-training quantization [@frantar2022gptq].

AWQ (https://arxiv.org/abs/2306.00978) uses activation-informed channel protection and an inference implementation [@lin2023awq].

SparseGPT (https://arxiv.org/abs/2301.00774) studies one-shot pruning [@frantar2023sparsegpt]. These are different representations, not matched fixed-rank baselines in the first wave. Their practical emphasis motivates recording actual checkpoint bytes and warmed end-to-end timings.

EoRA (https://arxiv.org/abs/2410.21271) compensates compressed models with eigenspace-informed low-rank branches [@liu2024eora].

SVDQuant (https://arxiv.org/abs/2411.05007) absorbs outliers in a high-precision low-rank branch while quantizing a residual branch, and identifies the runtime overhead of naive separate branch execution [@li2024svdquant]. SVDQuant's published scope is diffusion models. It supplies a system-design warning for future hybrid experiments here, not evidence of LLM quality or speedups.

CKA (https://proceedings.mlr.press/v97/kornblith19a.html) measures alignment

between representation similarities [@kornblith2019cka]. We use it as a secondary diagnostic beside the predeclared normalized residual squared error; neither representation similarity nor norm preservation substitutes for held-out NLL or logit KL. Temperature scaling (https://proceedings.mlr.press/v70/guo17a.html) is a validation-fitted calibration control [@guo2017calibration], adapted here to next-token probabilities. Improvement after temperature fitting would not by itself identify an LLM denoising mechanism.

Evaluation conventions can change apparent model rankings; the reproducibility guidance in Biderman et al. (https://arxiv.org/abs/2405.14782) motivates pinned assets, task formats and declared target masking [@biderman2024evaluation]. Pointer Sentinel Mixture Models (https://arxiv.org/abs/1609.07843) introduced WikiText [@merity2016wikitext]. Because WikiText has already informed this project, its present test evaluation is regression evidence. The later fresh confirmation samples a locked C4 shard, whose provenance is the T5 study (https://jmlr.org/papers/v21/20-074.html) [@raffel2020t5]. Neither its sampled confirmation nor the initial WikiText wave represents all text domains.

HellaSwag (https://aclanthology.org/P19-1472/) supplies a commonsense completion benchmark [@zellers2019hellaswag], and ARC (https://arxiv.org/abs/1803.05457) supplies science-question reasoning with an explicit Challenge subset [@clark2018arc]. Those downstream evaluations remain pending; no downstream accuracy claim follows from the readiness controls.

Verified bibliography and dates

The additive BibTeX file (paper_protocol_v2.bib) contains the 22 plan references, verified from primary metadata on 2026-10-13. Preprint entries use the first submission year, record the checked revision, and separately note a venue when the primary page identifies one. In particular SVD-LLM is a 2024 preprint with ICLR 2025 metadata, GPTQ is a 2022 preprint with ICLR 2023 metadata, AWQ is a 2023 preprint with MLSys 2024 metadata, and SVDQuant is a 2024 preprint with ICLR 2025 metadata. EoRA's current primary metadata identifies ICLR 2026 workshops, not main-conference publication. T5 is cited as its 2020 JMLR article, with its 2019 preprint date recorded separately. Unverified venues are omitted.

Author lists reflect the verified source version. The ASVD v5 author list and the EoRA v6 list differ from some earlier bibliographies; this file does not silently mix versions. Hsu et al.'s author spelling follows the conference PDF (Sung-En Chang); the arXiv metadata spells that name Sungen Chang. The direct OpenReview page returned a browser challenge, so metadata was cross-checked with the primary paper's indexed PDF and its arXiv record.

Recorded numerical controls

FP16 SVD readiness remains failed. Later BF16/FP8 variants are separate numerical cohorts, not repaired FP16 evidence. FP8 codes are decoded before BF16 multiplication; native tiny-GEMM support does not establish native model speed.

Wide-table continuation: identical row IDs identify the same original record across every panel. IDs restart for each source table.

Panel 1 of 2; match by Row ID.

Row ID	Run / state	Variant	NLL	PPL
R001	precision-readiness-20261014-060840-6db37daf / cancelled	teacher_bf16	3.4515495	31.5492399
R002	precision-readiness-20261014-060840-6db37daf / cancelled	copy_bf16	3.4515495	31.5492399
R003	precision-readiness-20261014-060840-6db37daf / cancelled	identity_bf16	3.4515495	31.5492399
R004	precision-readiness-20261014-060840-6db37daf / cancelled	svd_bf16	3.45265812	31.5842357
R005	precision-readiness-20261014-060840-6db37daf / cancelled	svd_fp32_compute	3.45158372	31.5503197
R006	precision-readiness-20261014-060840-6db37daf / cancelled	svd_fp8_e4m3	3.45542964	31.6718934
R007	precision-readiness-20261014-060840-6db37daf / cancelled	svd_fp8_e5m2	3.47055948	32.1547273
R008	readiness-20261013-164122-cdc919e9 / failed	teacher	3.45156254	31.5496515
R009	readiness-20261013-164122-cdc919e9 / failed	copied_teacher	3.45156254	31.5496515
R010	readiness-20261013-164407-bda5b4d9 / completed	teacher	3.45156254	31.5496515
R011	readiness-20261013-164407-bda5b4d9 / completed	copied_teacher	3.45156254	31.5496515
R012	readiness-20261013-164407-bda5b4d9 / completed	full_rank_factorized	3.45156254	31.5496515
R013	svd-diagnosis-20261013-165146-c3671715 / completed	teacher	3.45156254	31.5496515
R014	svd-diagnosis-20261013-165146-c3671715 / completed	dense_svd_fp32	3.4516607	31.5527485
R015	svd-diagnosis-20261013-165146-c3671715 / completed	dense_svd_fp16	3.45170481	31.5541402
R016	svd-diagnosis-20261013-165146-c3671715 / completed	unbalanced_fp16	3.45186016	31.5590426
R017	svd-diagnosis-20261013-165146-c3671715 / completed	balanced_fp16	3.45184835	31.5586697
R018	svd-diagnosis-20261013-165146-c3671715 / completed	float32_execution	3.4517713	31.5562382

Panel 2 of 2; match by Row ID.

Row ID	Run / state	Maximum probe error	Probe gate	NLL gate
R001	precision-readiness-20261014-060840-6db37daf / cancelled	0	True	True
R002	precision-readiness-20261014-060840-6db37daf / cancelled	0	True	True
R003	precision-readiness-20261014-060840-6db37daf / cancelled	0	True	True
R004	precision-readiness-20261014-060840-6db37daf / cancelled	1.0625	False	False
R005	precision-readiness-20261014-060840-6db37daf / cancelled	1.15625	False	True
R006	precision-readiness-20261014-060840-6db37daf / cancelled	9.73046875	False	False
R007	precision-readiness-20261014-060840-6db37daf / cancelled	7.8671875	False	False

Row ID	Run / state	Maximum probe error	Probe gate	NLL gate
R008	readiness-20261013-164122-cdc919e9 / failed	0	True	unrecorded
R009	readiness-20261013-164122-cdc919e9 / failed	0	True	unrecorded
R010	readiness-20261013-164407-bda5b4d9 / completed	0	True	unrecorded
R011	readiness-20261013-164407-bda5b4d9 / completed	0	True	unrecorded
R012	readiness-20261013-164407-bda5b4d9 / completed	0	True	unrecorded
R013	svd-diagnosis-20261013-165146-c3671715 / completed	unmeasured	unrecorded	unrecorded
R014	svd-diagnosis-20261013-165146-c3671715 / completed	0.1171875	True	True
R015	svd-diagnosis-20261013-165146-c3671715 / completed	0.140625	False	True
R016	svd-diagnosis-20261013-165146-c3671715 / completed	0.15234375	False	True
R017	svd-diagnosis-20261013-165146-c3671715 / completed	0.181640625	False	True
R018	svd-diagnosis-20261013-165146-c3671715 / completed	0.1171875	True	True

Validation regularization controls

Every row identifies its attempt and scope. Mean ± sample standard deviation is across available calibration seeds; a single seed has no estimated seed standard deviation. Incomplete attempts and partial seed coverage do not constitute a full cohort.

Wide-table continuation: identical row IDs identify the same original record across every panel. IDs restart for each source table.

Panel 1 of 3; match by Row ID.

Row ID	Run / state	Specification	Split / group / rank scope	Seeds
R001	regularization-20261013-171941-804f4c2e / completed	{"method": "svd"}	{}	[11]
R002	regularization-20261013-171941-804f4c2e / completed	{"beta": 0.1, "method": "fullrank_ridge", "ridge_fraction": 0.01}	{}	[11]
R003	regularization-20261013-171941-804f4c2e / completed	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{}	[11]
R004	regularization-20261013-171941-804f4c2e / completed	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{}	[11]
R005	regularization-20261013-171941-804f4c2e / completed	{"beta": 0.1, "method": "anchored", "ridge_fraction": 0.0}	{}	[11]
R006	regularization-20261013-171941-804f4c2e / completed	{"beta": 0.3, "method": "anchored", "ridge_fraction": 0.0}	{}	[11]
R007	regularization-20261013-171941-804f4c2e / completed	{"beta": 1.0, "method": "anchored", "ridge_fraction": 0.0}	{}	[11]
R008	regularization-20261013-171941-804f4c2e / completed	{"beta": 0.1, "method": "legacy_ridge", "ridge_fraction": 0.0}	{}	[11]
R009	regularization-20261013-171941-804f4c2e / completed	{"beta": 0.1, "method": "legacy_ridge", "ridge_fraction": 0.001}	{}	[11]
R010	regularization-20261013-171941-804f4c2e / completed	{"beta": 0.1, "method": "legacy_ridge", "ridge_fraction": 0.01}	{}	[11]

Row ID	Run / state	Specification	Split / group / rank scope	Seeds
R011	regularization-20261013-171941-804f4c2e / completed	{"method": "asvd"}	{}	[11]

Panel 2 of 3; match by Row ID.

Row ID	Run / state	NLL mean	NLL seed SD	Drift mean
R001	regularization-20261013-171941-804f4c2e / completed	5.14486078	unmeasured	unmeasured
R002	regularization-20261013-171941-804f4c2e / completed	3.49923448	unmeasured	unmeasured
R003	regularization-20261013-171941-804f4c2e / completed	3.60078899	unmeasured	unmeasured
R004	regularization-20261013-171941-804f4c2e / completed	3.59662239	unmeasured	unmeasured
R005	regularization-20261013-171941-804f4c2e / completed	3.59675176	unmeasured	unmeasured
R006	regularization-20261013-171941-804f4c2e / completed	3.607465	unmeasured	unmeasured
R007	regularization-20261013-171941-804f4c2e / completed	3.61579886	unmeasured	unmeasured
R008	regularization-20261013-171941-804f4c2e / completed	3.59685438	unmeasured	unmeasured
R009	regularization-20261013-171941-804f4c2e / completed	3.59727845	unmeasured	unmeasured
R010	regularization-20261013-171941-804f4c2e / completed	3.60682971	unmeasured	unmeasured
R011	regularization-20261013-171941-804f4c2e / completed	3.73607467	unmeasured	unmeasured

Panel 3 of 3; match by Row ID.

Row ID	Run / state	Drift seed SD
R001	regularization-20261013-171941-804f4c2e / completed	unmeasured
R002	regularization-20261013-171941-804f4c2e / completed	unmeasured
R003	regularization-20261013-171941-804f4c2e / completed	unmeasured
R004	regularization-20261013-171941-804f4c2e / completed	unmeasured
R005	regularization-20261013-171941-804f4c2e / completed	unmeasured
R006	regularization-20261013-171941-804f4c2e / completed	unmeasured
R007	regularization-20261013-171941-804f4c2e / completed	unmeasured
R008	regularization-20261013-171941-804f4c2e / completed	unmeasured
R009	regularization-20261013-171941-804f4c2e / completed	unmeasured
R010	regularization-20261013-171941-804f4c2e / completed	unmeasured
R011	regularization-20261013-171941-804f4c2e / completed	unmeasured

Gemma full-stack component comparisons

Every row identifies its attempt and scope. Mean ± sample standard deviation is across available calibration seeds; a single seed has no estimated seed standard deviation. Incomplete attempts and partial seed coverage do not constitute a full cohort.

Wide-table continuation: identical row IDs identify the same original record across every panel. IDs restart for each source table.

Panel 1 of 3; match by Row ID.

Row ID	Run / state	Specification	Split / group / rank scope	Seeds
R001	baselines-20261013-175255-6a00d4e8 / completed	{"method": "svd"}	{}	[11]

Row ID	Run / state	Specification	Split / group / rank scope	Seeds
R002	baselines-20261013-175255-6a00d4e8 / completed	{"method": "asvd"}	{}	[11, 23, 37]
R003	baselines-20261013-175255-6a00d4e8 / completed	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{}	[11, 23, 37]
R004	baselines-20261013-175255-6a00d4e8 / completed	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{}	[11, 23, 37]
R005	baselines-20261013-175255-6a00d4e8 / completed	{"beta": 0.1, "method": "legacy_ridge", "ridge_fraction": 0.01}	{}	[11, 23, 37]

Panel 2 of 3; match by Row ID.

Row ID	Run / state	NLL mean	NLL seed SD	Drift mean
R001	baselines-20261013-175255-6a00d4e8 / completed	5.08868112	unmeasured	0.0742475831
R002	baselines-20261013-175255-6a00d4e8 / completed	3.69446606	0.00241956806	0.0309717453
R003	baselines-20261013-175255-6a00d4e8 / completed	3.57281383	0.0118191598	0.0106375166
R004	baselines-20261013-175255-6a00d4e8 / completed	3.56592113	0.00957592299	0.0105470011
R005	baselines-20261013-175255-6a00d4e8 / completed	3.57903427	0.00852077408	0.0118359709

Panel 3 of 3; match by Row ID.

Row ID	Run / state	Drift seed SD
R001	baselines-20261013-175255-6a00d4e8 / completed	unmeasured
R002	baselines-20261013-175255-6a00d4e8 / completed	0.00019731531
R003	baselines-20261013-175255-6a00d4e8 / completed	0.000382136395
R004	baselines-20261013-175255-6a00d4e8 / completed	0.000492296849
R005	baselines-20261013-175255-6a00d4e8 / completed	0.000138552104

Composition and group heterogeneity

Every row identifies its attempt and scope. Mean ± sample standard deviation is across available calibration seeds; a single seed has no estimated seed standard deviation. Incomplete attempts and partial seed coverage do not constitute a full cohort.

Wide-table continuation: identical row IDs identify the same original record across every panel. IDs restart for each source table.

Panel 1 of 3; match by Row ID.

Row ID	Run / state	Specification	Split / group / rank scope	Seeds
R001	composition-20261013-185512-2bffd477 / interrupted	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [0], "rank": 384, "split": "test"}	[11, 23, 37]
R002	composition-20261013-185512-2bffd477 / interrupted	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [0], "rank": 384, "split": "test"}	[11, 23, 37]
R003	composition-20261013-185512-2bffd477 / interrupted	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [12], "rank": 384, "split": "test"}	[11, 23, 37]
R004	composition-20261013-185512-2bffd477 / interrupted	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [12], "rank": 384, "split": "test"}	[11, 23, 37]
R005	composition-20261013-185512-2bffd477 / interrupted	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [25], "rank": 384, "split": "test"}	[11, 23, 37]

Row ID	Run / state	Specification	Split / group / rank scope	Seeds
R006	composition-20261013-185512-2bffd477 / interrupted	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [25], "rank": 384, "split": "test"}	[11, 23, 37]
R007	composition-20261013-185512-2bffd477 / interrupted	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [0, 1], "rank": 384, "split": "test"}	[11, 23, 37]
R008	composition-20261013-185512-2bffd477 / interrupted	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [0, 1], "rank": 384, "split": "test"}	[11, 23, 37]
R009	composition-20261013-185512-2bffd477 / interrupted	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [12, 13], "rank": 384, "split": "test"}	[11, 23, 37]
R010	composition-20261013-185512-2bffd477 / interrupted	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [12, 13], "rank": 384, "split": "test"}	[11, 23, 37]
R011	composition-20261013-185512-2bffd477 / interrupted	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [24, 25], "rank": 384, "split": "test"}	[11, 23, 37]
R012	composition-20261013-185512-2bffd477 / interrupted	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [24, 25], "rank": 384, "split": "test"}	[11]
R013	composition-20261014-032120-486d799f / interrupted	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [0], "rank": 384, "split": "test"}	[11, 23, 37]
R014	composition-20261014-032120-486d799f / interrupted	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [0], "rank": 384, "split": "test"}	[11, 23, 37]
R015	composition-20261014-032120-486d799f / interrupted	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [12], "rank": 384, "split": "test"}	[11, 23, 37]
R016	composition-20261014-032120-486d799f / interrupted	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [12], "rank": 384, "split": "test"}	[11, 23, 37]
R017	composition-20261014-032120-486d799f / interrupted	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [25], "rank": 384, "split": "test"}	[11, 23, 37]
R018	composition-20261014-032120-486d799f / interrupted	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [25], "rank": 384, "split": "test"}	[11, 23, 37]
R019	composition-20261014-032120-486d799f / interrupted	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [0, 1], "rank": 384, "split": "test"}	[11, 23, 37]
R020	composition-20261014-032120-486d799f / interrupted	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [0, 1], "rank": 384, "split": "test"}	[11, 23, 37]
R021	composition-20261014-032120-486d799f / interrupted	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [12, 13], "rank": 384, "split": "test"}	[11, 23, 37]
R022	composition-20261014-032120-486d799f / interrupted	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [12, 13], "rank": 384, "split": "test"}	[11, 23, 37]
R023	composition-20261014-032120-486d799f / interrupted	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [24, 25], "rank": 384, "split": "test"}	[11, 23, 37]
R024	composition-20261014-032120-486d799f / interrupted	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [24, 25], "rank": 384, "split": "test"}	[11, 23, 37]
R025	composition-20261014-032120-486d799f / interrupted	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [0, 1, 2, 3], "rank": 384, "split": "test"}	[11, 23, 37]
R026	composition-20261014-032120-486d799f / interrupted	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [0, 1, 2, 3], "rank": 384, "split": "test"}	[11, 23, 37]

Row ID	Run / state	Specification	Split / group / rank scope	Seeds
R027	composition-20261014-032120-486d799f / interrupted	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [11, 12, 13, 14], "rank": 384, "split": "test"}	[11, 23, 37]
R028	composition-20261014-032120-486d799f / interrupted	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [11, 12, 13, 14], "rank": 384, "split": "test"}	[11, 23]
R029	composition-20261014-054423-dbd821ff / completed	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [0], "rank": 384, "split": "test"}	[11, 23, 37]
R030	composition-20261014-054423-dbd821ff / completed	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [0], "rank": 384, "split": "test"}	[11, 23, 37]
R031	composition-20261014-054423-dbd821ff / completed	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [12], "rank": 384, "split": "test"}	[11, 23, 37]
R032	composition-20261014-054423-dbd821ff / completed	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [12], "rank": 384, "split": "test"}	[11, 23, 37]
R033	composition-20261014-054423-dbd821ff / completed	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [25], "rank": 384, "split": "test"}	[11, 23, 37]
R034	composition-20261014-054423-dbd821ff / completed	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [25], "rank": 384, "split": "test"}	[11, 23, 37]
R035	composition-20261014-054423-dbd821ff / completed	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [0, 1], "rank": 384, "split": "test"}	[11, 23, 37]
R036	composition-20261014-054423-dbd821ff / completed	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [0, 1], "rank": 384, "split": "test"}	[11, 23, 37]
R037	composition-20261014-054423-dbd821ff / completed	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [12, 13], "rank": 384, "split": "test"}	[11, 23, 37]
R038	composition-20261014-054423-dbd821ff / completed	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [12, 13], "rank": 384, "split": "test"}	[11, 23, 37]
R039	composition-20261014-054423-dbd821ff / completed	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [24, 25], "rank": 384, "split": "test"}	[11, 23, 37]
R040	composition-20261014-054423-dbd821ff / completed	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [24, 25], "rank": 384, "split": "test"}	[11, 23, 37]
R041	composition-20261014-054423-dbd821ff / completed	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [0, 1, 2, 3], "rank": 384, "split": "test"}	[11, 23, 37]
R042	composition-20261014-054423-dbd821ff / completed	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [0, 1, 2, 3], "rank": 384, "split": "test"}	[11, 23, 37]
R043	composition-20261014-054423-dbd821ff / completed	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [11, 12, 13, 14], "rank": 384, "split": "test"}	[11, 23, 37]
R044	composition-20261014-054423-dbd821ff / completed	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [11, 12, 13, 14], "rank": 384, "split": "test"}	[11, 23, 37]
R045	composition-20261014-054423-dbd821ff / completed	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [22, 23, 24, 25], "rank": 384, "split": "test"}	[11, 23, 37]
R046	composition-20261014-054423-dbd821ff / completed	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [22, 23, 24, 25], "rank": 384, "split": "test"}	[11, 23, 37]
R047	composition-20261014-054423-dbd821ff / completed	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25], "rank": 384, "split": "test"}	[11, 23, 37]

Row ID	Run / state	Specification	Split / group / rank scope	Seeds
R048	composition-20261014-054423-dbd821ff / completed	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25], "rank": 384, "split": "test"}	[11, 23, 37]

Panel 2 of 3; match by Row ID.

Row ID	Run / state	NLL mean	NLL seed SD	Drift mean
R001	composition-20261013-185512-2bffd477 / interrupted	3.45206785	0.000410778252	0.00207792164
R002	composition-20261013-185512-2bffd477 / interrupted	3.45201942	0.000409256997	0.00207452887
R003	composition-20261013-185512-2bffd477 / interrupted	3.45825421	0.00132397251	8.76745161e-05
R004	composition-20261013-185512-2bffd477 / interrupted	3.4579111	0.00143757277	8.91802058e-05
R005	composition-20261013-185512-2bffd477 / interrupted	3.45142891	0.000329325004	1.69831209e-06
R006	composition-20261013-185512-2bffd477 / interrupted	3.45122641	0.000131137763	2.023157e-06
R007	composition-20261013-185512-2bffd477 / interrupted	3.45442908	0.00205462111	0.00365644585
R008	composition-20261013-185512-2bffd477 / interrupted	3.45438131	0.00214728679	0.00356760921
R009	composition-20261013-185512-2bffd477 / interrupted	3.46353003	0.00115146509	0.000141608504
R010	composition-20261013-185512-2bffd477 / interrupted	3.46353582	0.00129669463	0.000144554107
R011	composition-20261013-185512-2bffd477 / interrupted	3.45212529	0.000451146203	1.23266207e-05
R012	composition-20261013-185512-2bffd477 / interrupted	3.45195513	unmeasured	1.21042136e-05
R013	composition-20261014-032120-486d799f / interrupted	3.45206785	0.000410778252	0.00207792164
R014	composition-20261014-032120-486d799f / interrupted	3.45201942	0.000409256997	0.00207452887
R015	composition-20261014-032120-486d799f / interrupted	3.45825421	0.00132397251	8.76745161e-05
R016	composition-20261014-032120-486d799f / interrupted	3.4579111	0.00143757277	8.91802058e-05
R017	composition-20261014-032120-486d799f / interrupted	3.45142891	0.000329325004	1.69831209e-06
R018	composition-20261014-032120-486d799f / interrupted	3.45122641	0.000131137763	2.023157e-06
R019	composition-20261014-032120-486d799f / interrupted	3.45442908	0.00205462111	0.00365644585
R020	composition-20261014-032120-486d799f / interrupted	3.45438131	0.00214728679	0.00356760921
R021	composition-20261014-032120-486d799f / interrupted	3.46353003	0.00115146509	0.000141608504
R022	composition-20261014-032120-486d799f / interrupted	3.46353582	0.00129669463	0.000144554107
R023	composition-20261014-032120-486d799f / interrupted	3.45212529	0.000451146203	1.23266207e-05
R024	composition-20261014-032120-486d799f / interrupted	3.45194198	0.000101102007	1.22970593e-05
R025	composition-20261014-032120-486d799f / interrupted	3.46224947	0.000734496385	0.0061389891

Row ID	Run / state	NLL mean	NLL seed SD	Drift mean
R026	composition-20261014-032120-486d799f / interrupted	3.4619553	0.00102918121	0.00600512161
R027	composition-20261014-032120-486d799f / interrupted	3.47310172	0.000470679639	0.00038058521
R028	composition-20261014-032120-486d799f / interrupted	3.4765793	1.58332576e-05	0.000438405635
R029	composition-20261014-054423-dbd821ff / completed	3.45206785	0.000410778252	0.00207792164
R030	composition-20261014-054423-dbd821ff / completed	3.45201942	0.000409256997	0.00207452887
R031	composition-20261014-054423-dbd821ff / completed	3.45825421	0.00132397251	8.76745161e-05
R032	composition-20261014-054423-dbd821ff / completed	3.4579111	0.00143757277	8.91802058e-05
R033	composition-20261014-054423-dbd821ff / completed	3.45142891	0.000329325004	1.69831209e-06
R034	composition-20261014-054423-dbd821ff / completed	3.45122641	0.000131137763	2.023157e-06
R035	composition-20261014-054423-dbd821ff / completed	3.45442908	0.00205462111	0.00365644585
R036	composition-20261014-054423-dbd821ff / completed	3.45438131	0.00214728679	0.00356760921
R037	composition-20261014-054423-dbd821ff / completed	3.46353003	0.00115146509	0.000141608504
R038	composition-20261014-054423-dbd821ff / completed	3.46353582	0.00129669463	0.000144554107
R039	composition-20261014-054423-dbd821ff / completed	3.45212529	0.000451146203	1.23266207e-05
R040	composition-20261014-054423-dbd821ff / completed	3.45194198	0.000101102007	1.22970593e-05
R041	composition-20261014-054423-dbd821ff / completed	3.46224947	0.000734496385	0.0061389891
R042	composition-20261014-054423-dbd821ff / completed	3.4619553	0.00102918121	0.00600512161
R043	composition-20261014-054423-dbd821ff / completed	3.47310172	0.000470679639	0.00038058521
R044	composition-20261014-054423-dbd821ff / completed	3.47582544	0.00130576482	0.000449140822
R045	composition-20261014-054423-dbd821ff / completed	3.45396348	0.00136122732	0.000123680895
R046	composition-20261014-054423-dbd821ff / completed	3.4529696	0.000385430754	0.00011315281
R047	composition-20261014-054423-dbd821ff / completed	3.57281383	0.0118191598	0.0106375166
R048	composition-20261014-054423-dbd821ff / completed	3.56592113	0.00957592299	0.0105470011

Panel 3 of 3; match by Row ID.

Row ID	Run / state	Drift seed SD
R001	composition-20261013-185512-2bffd477 / interrupted	0.00017472334
R002	composition-20261013-185512-2bffd477 / interrupted	0.000178507413
R003	composition-20261013-185512-2bffd477 / interrupted	2.4463997e-06
R004	composition-20261013-185512-2bffd477 / interrupted	2.24916162e-06
R005	composition-20261013-185512-2bffd477 / interrupted	9.05406025e-08
R006	composition-20261013-185512-2bffd477 / interrupted	9.42299447e-08

Row ID	Run / state	Drift seed SD
R007	composition-20261013-185512-2bffd477 / interrupted	0.000137627131
R008	composition-20261013-185512-2bffd477 / interrupted	0.000238304358
R009	composition-20261013-185512-2bffd477 / interrupted	4.06719991e-06
R010	composition-20261013-185512-2bffd477 / interrupted	4.09979657e-06
R011	composition-20261013-185512-2bffd477 / interrupted	3.68456941e-07
R012	composition-20261013-185512-2bffd477 / interrupted	unmeasured
R013	composition-20261014-032120-486d799f / interrupted	0.00017472334
R014	composition-20261014-032120-486d799f / interrupted	0.000178507413
R015	composition-20261014-032120-486d799f / interrupted	2.4463997e-06
R016	composition-20261014-032120-486d799f / interrupted	2.24916162e-06
R017	composition-20261014-032120-486d799f / interrupted	9.05406025e-08
R018	composition-20261014-032120-486d799f / interrupted	9.42299447e-08
R019	composition-20261014-032120-486d799f / interrupted	0.000137627131
R020	composition-20261014-032120-486d799f / interrupted	0.000238304358
R021	composition-20261014-032120-486d799f / interrupted	4.06719991e-06
R022	composition-20261014-032120-486d799f / interrupted	4.09979657e-06
R023	composition-20261014-032120-486d799f / interrupted	3.68456941e-07
R024	composition-20261014-032120-486d799f / interrupted	3.91576035e-07
R025	composition-20261014-032120-486d799f / interrupted	0.000142663396
R026	composition-20261014-032120-486d799f / interrupted	0.000161076046
R027	composition-20261014-032120-486d799f / interrupted	1.72440575e-05
R028	composition-20261014-032120-486d799f / interrupted	3.31525237e-07
R029	composition-20261014-054423-dbd821ff / completed	0.00017472334
R030	composition-20261014-054423-dbd821ff / completed	0.000178507413
R031	composition-20261014-054423-dbd821ff / completed	2.4463997e-06
R032	composition-20261014-054423-dbd821ff / completed	2.24916162e-06
R033	composition-20261014-054423-dbd821ff / completed	9.05406025e-08
R034	composition-20261014-054423-dbd821ff / completed	9.42299447e-08
R035	composition-20261014-054423-dbd821ff / completed	0.000137627131
R036	composition-20261014-054423-dbd821ff / completed	0.000238304358
R037	composition-20261014-054423-dbd821ff / completed	4.06719991e-06
R038	composition-20261014-054423-dbd821ff / completed	4.09979657e-06
R039	composition-20261014-054423-dbd821ff / completed	3.68456941e-07
R040	composition-20261014-054423-dbd821ff / completed	3.91576035e-07
R041	composition-20261014-054423-dbd821ff / completed	0.000142663396
R042	composition-20261014-054423-dbd821ff / completed	0.000161076046
R043	composition-20261014-054423-dbd821ff / completed	1.72440575e-05
R044	composition-20261014-054423-dbd821ff / completed	1.85953677e-05
R045	composition-20261014-054423-dbd821ff / completed	1.21816693e-06
R046	composition-20261014-054423-dbd821ff / completed	2.34008283e-06
R047	composition-20261014-054423-dbd821ff / completed	0.000382136395
R048	composition-20261014-054423-dbd821ff / completed	0.000492296849

Group-dependent outcomes must not be pooled into a universal-improvement claim. A lower full-stack NLL with unchanged/worse residual drift does not satisfy the predeclared joint gate. Three paired seeds with both higher NLL and higher drift weaken the candidate only at that measured scope.

Amended hybrid architecture and rank-frontier comparisons

Every row identifies its attempt and scope. Mean ± sample standard deviation is across available calibration seeds; a single seed has no estimated seed standard deviation. Incomplete attempts and partial seed coverage do not constitute a full cohort.

Wide-table continuation: identical row IDs identify the same original record across every panel. IDs restart for each source table.

Panel 1 of 3; match by Row ID.

Row ID	Run / state	Specification	Split / group / rank scope	Seeds
R001	architecture-replication-20261014-082807-5c73f669 / running	{"method": "teacher"}	{"rank": null, "rank_divisor": null, "recipe_ranks": [null, null, null, null, null, null], "split": "test"}	[11]
R002	architecture-replication-20261014-082807-5c73f669 / running	{"method": "svd"}	{"rank": 483, "rank_divisor": null, "recipe_ranks": [483, 483, 483, 483, 483, 483], "split": "test"}	[11]
R003	architecture-replication-20261014-082807-5c73f669 / running	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"rank": 483, "rank_divisor": null, "recipe_ranks": [483, 483, 483, 483, 483, 483], "split": "test"}	[11, 23]
R004	architecture-replication-20261014-082807-5c73f669 / running	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"rank": 483, "rank_divisor": null, "recipe_ranks": [483, 483, 483, 483, 483, 483], "split": "test"}	[11]

Panel 2 of 3; match by Row ID.

Row ID	Run / state	NLL mean	NLL seed SD	Drift mean
R001	architecture-replication-20261014-082807-5c73f669 / running	2.91722043	unmeasured	0
R002	architecture-replication-20261014-082807-5c73f669 / running	3.18099289	unmeasured	0.0616629847
R003	architecture-replication-20261014-082807-5c73f669 / running	2.93525973	0.00043802431	0.0061133559
R004	architecture-replication-20261014-082807-5c73f669 / running	2.94489166	unmeasured	0.0127864543

Panel 3 of 3; match by Row ID.

Row ID	Run / state	Drift seed SD
R001	architecture-replication-20261014-082807-5c73f669 / running	unmeasured
R002	architecture-replication-20261014-082807-5c73f669 / running	unmeasured
R003	architecture-replication-20261014-082807-5c73f669 / running	7.84021734e-05
R004	architecture-replication-20261014-082807-5c73f669 / running	unmeasured

Practical costs and calibration

Whole-model checkpoint bytes include archive overhead; factor payload size and whole-model savings are separate quantities. Placements and precision must agree before comparing timings. Below are measured records only; missing conditions remain unmeasured.

Wide-table continuation: identical row IDs identify the same original record across every panel. IDs restart for each source table.

Panel 1 of 5; match by Row ID.

Row ID	Run / state	Label	Seed	Bits
R001	Unmeasured	pending	pending	pending

Panel 2 of 5; match by Row ID.

Row ID	Run / state	Whole bytes	Factor bytes	Raw NLL
R001	Unmeasured	unmeasured	unmeasured	unmeasured

Panel 3 of 5; match by Row ID.

Row ID	Run / state	Scaled NLL	Temperature	Mode
R001	Unmeasured	unmeasured	unmeasured	pending

Panel 4 of 5; match by Row ID.

Row ID	Run / state	Prefill	Median s	p95 s
R001	Unmeasured	pending	unmeasured	unmeasured

Panel 5 of 5; match by Row ID.

Row ID	Run / state	Allocated bytes	Reserved bytes
R001	Unmeasured	unmeasured	unmeasured

Statistical interpretation

NLL differences use 2,000 paired bootstrap resamples of fixed contiguous 4,096-target blocks. The final partial block remains in full-test NLL but is excluded from fixed-block resampling. Individual dependent tokens are not bootstrapped. Per-seed confidence intervals, exact excluded counts, matched target checks and all scoped candidate-minus-baseline differences are in report.json and report.md. These are evaluation-block intervals, not seed intervals.

Reproducibility and incomplete-run appendix

The accompanying report.json embeds each retained contract, source hashes, package/device versions, asset revisions, calibration/split hashes, target recipes, run-specific command, raw per-seed metrics, integrity checks, artifact paths and all failed/interrupted attempts. Read errors are retained explicitly; promotion is disabled.

Stage	Recorded charged wall seconds	Declared cap seconds	Unknown-cost attempts
readiness	1381.392	3600	[]
regularization	1589.148	14400	[]
baselines	3736.553	25200	[]
composition	12825.574	14400	[]
architecture	0.000	14400	['architecture-replication-20261014-082807-5c73f669']
practical	0.000	14400	[]

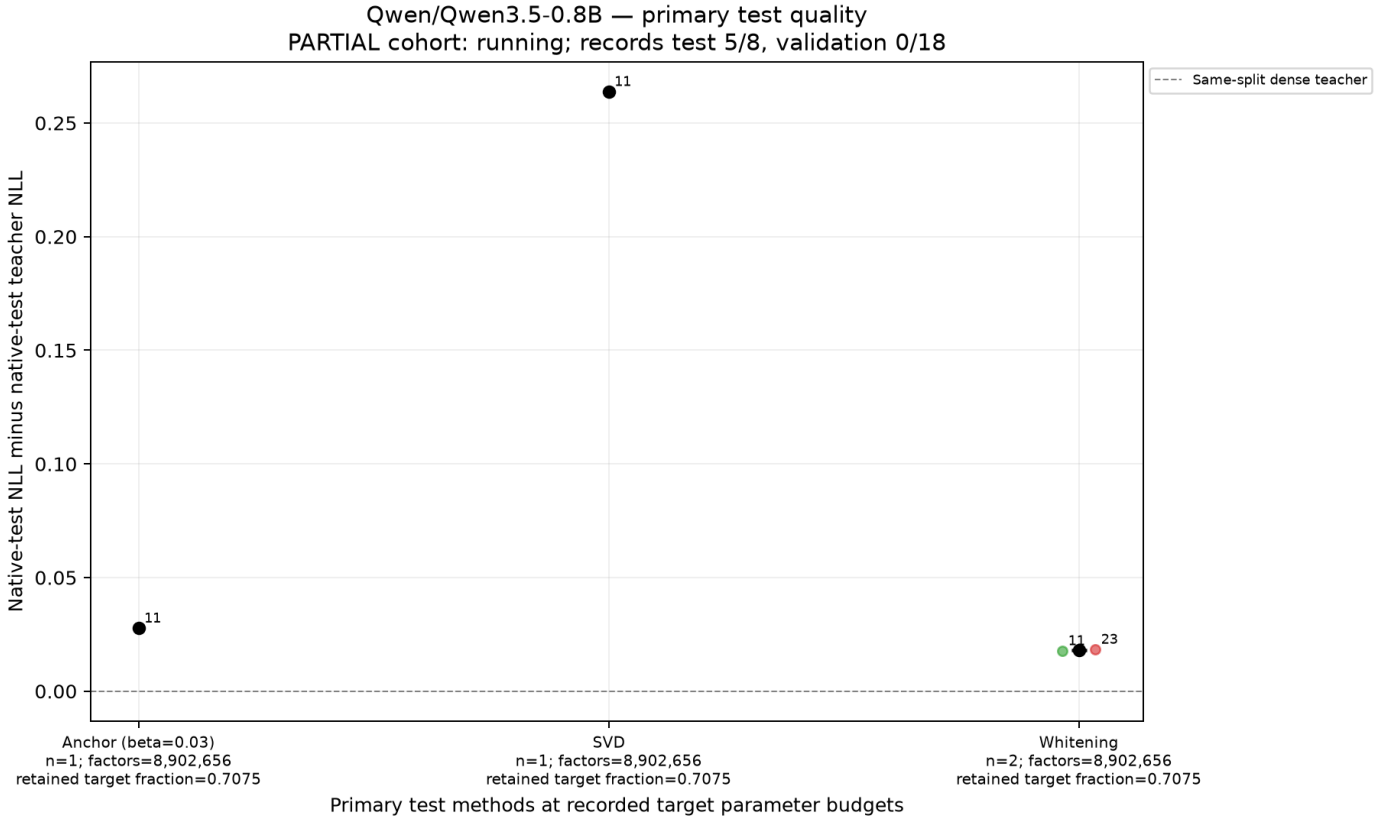
Elapsed-wall-time charges include failed/retried work and may include reboot downtime. The first reboot has an unbounded unlogged tail; no exact GPU-hour compliance claim.

Conclusion

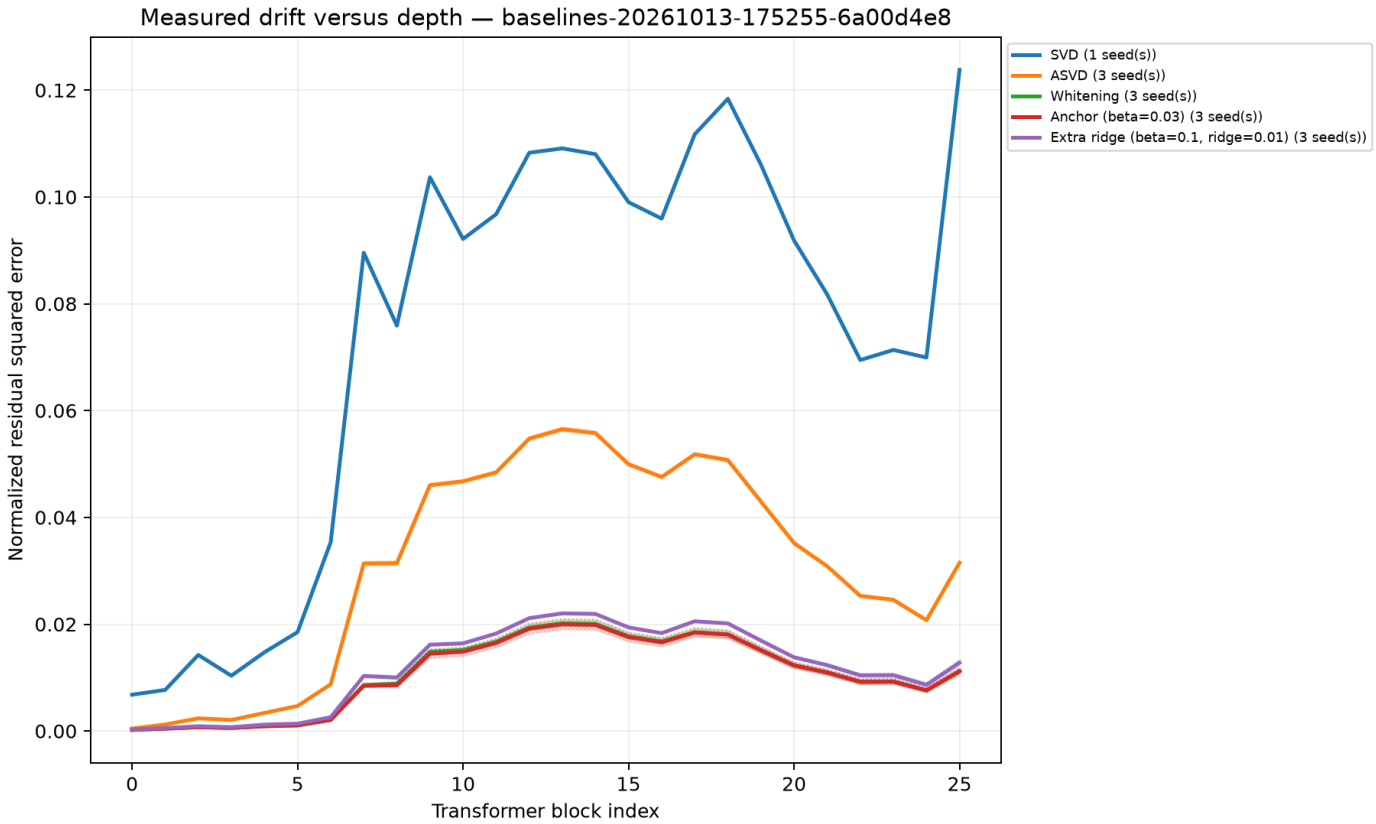
No established practical or universal compression advantage; retained within-model findings remain exploratory.

Complete negative or mixed comparisons are retained as bounded evidence. Unmeasured downstream tasks, fresh-corpus confirmation, absent seeds or hardware conditions are not implied by dispatcher completion.

Measured scientific figures



Qwen/Qwen3.5-0.8B: primary native-test teacher-relative NLL for PARTIAL cohort: running; records test 5/8, validation 0/18; individual paired seed points, mean and sample seed SD when at least two seeds exist. Actual target factor counts/fractions shown. Strict promotion remains disabled.



Gemma Stage C residual drift: individual seeds (dotted), mean (solid), sample seed SD (shaded). Single retained cohort; strict FP16 readiness remains failed.

Figure inputs, scope, source hashes and unavailable measurements are recorded in figures/provenance.json.

Review export formats

Standalone LaTeX source and a separately generated ReportLab PDF are review artifacts. Their preparation status and source hashes are recorded in export-status.json. The ReportLab PDF is not compiled from LaTeX; successful TeX compilation requires a separate compiler diagnostic.

Paired-block uncertainty appendix

Intervals are candidate minus the same-split baseline, using evaluation blocks rather than individual tokens. Partial-tail targets remain included in full NLL and are excluded from fixed-block intervals. These are not calibration-seed intervals.

architecture-replication-20261014-082807-5c73f669

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"method": "svd"}, "scope": {"rank": 483, "rank_divisor": null, "split": "test", "target_recipes": [{"block_index": 3, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.3.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}, {"block_index": 7, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.7.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}, {"block_index": 11, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.11.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}, {"block_index": 15, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.15.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}, {"block_index": 19, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.19.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}, {"block_index": 23, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.23.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}]}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	0.24604289213573946	0.24615020975613225	[0.24006466116265526, 0.25237916880828776]	72/2000	2280

architecture-replication-20261014-082807-5c73f669

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"rank": 483, "rank_divisor": null, "split": "test", "target_recipes": [{"block_index": 3, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.3.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}, {"block_index": 7, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.7.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}, {"block_index": 11, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.11.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}, {"block_index": 15, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.15.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}, {"block_index": 19, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.19.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}, {"block_index": 23, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.23.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}]}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	0.009941660681098283	0.010034525719555844	[0.00827384469021905, 0.01189700522595723]	72/2000	2280

baselines-20261013-175255-6a00d4e8

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"method": "svd"}, "scope": {}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	1.5259270359038637	1.526963313415228	[1.4943535833878134, 1.5596674649359294]	71/2000	1465

baselines-20261013-175255-6a00d4e8

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"method": "asvd"}, "scope": {}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	0.13435744983933437	0.13448046045923576	[0.12208941117886445, 0.14684303527631282]	71/2000	1465
23	0.12406477628810314	0.12440374314135588	[0.1140738804768457, 0.13439828040507174]	71/2000	1465
37	0.10653445125745864	0.10667353359540374	[0.09858797073867313, 0.11492986771691276]	71/2000	1465

baselines-20261013-175255-6a00d4e8

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	-0.0031192516557841613	-0.003069222128221489	[-0.005598433325437209, -0.0005771062229211223	71/2000	1465

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
			]		
23	-0.00867016563638367	-0.008560964606249673	[-0.01236161516313488, -0.004879390368432393]	71/2000	1465
37	-0.008888696495308254	-0.008889474302069626	[-0.011564302129920533, -0.006193025270745394]	71/2000	1465

baselines-20261013-175255-6a00d4e8

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.1, "method": "legacy_ridge", "ridge_fraction": 0.01}, "scope": {}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	0.007341743295384272	0.007358043732914079	[0.003578168563012017, 0.011322591469446464]	71/2000	1465
23	0.010085729923213327	0.010195171973247596	[0.005499654363179419, 0.01504279117529553]	71/2000	1465
37	0.0012338436455246004	0.0012706044011388724	[-0.002438464246056019, 0.00520269695332208]	71/2000	1465

composition-20261013-185512-2bffd477

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [0], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	-4.60899341388781e-05	-4.0872599009493115e-05	[-0.000217734233338580 93, 0.0001685818120544 4938]	71/2000	1465
23	-3.271375526325137e-05	-3.199588131774163e-05	[-0.000174187935932211 99, 0.0001136384687997568]	71/2000	1465
37	-6.647979225569334e-05	-7.095429546745745e-05	[-0.000291885207220884 4, 9.73636943746164e-05]	71/2000	1465

composition-20261013-185512-2bffd477

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [12], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	-0.0008545763004039009	-0.0008561104717573092	[-0.001150257172183655 8, -0.00056394938574460 14]	71/2000	1465
23	-9.84808364817539e-05	-9.306537422449584e-05	[-0.000375805062010882 6, 0.00017279865033056 432]	71/2000	1465
37	-7.626569138096745e-05	-7.009637664775361e-05	[-0.000360900941821140 16, 0.00023415362379323]	71/2000	1465

composition-20261013-185512-2bffd477

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [25], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	0.0001729304994845724 8	0.0001719993236095868 5	[4.463885584017398e-05, 0.0002936832496752188]	71/2000	1465
23	-0.0007079791885041549	-0.000699445138327318	[-0.000889761639764939 4, -0.000521697610992854]	71/2000	1465
37	-7.243953146688717e-05	-6.619463365792254e-05	[-0.000212401684606083 8, 7.996392602977384e-05]	71/2000	1465

composition-20261013-185512-2bffd477

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [0, 1], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	-0.0002764577707105269	-0.000305017713926657	[-0.000801674596175575 1, 0.00016777266329458292]	71/2000	1465
23	0.0002748313234466563 5	0.0002560327026930727	[-0.000328662642760605 2, 0.0008294795855742868]	71/2000	1465
37	-0.0001416761346267915 2	-0.0001526383887316477 5	[-0.000643328925356078 7, 0.0003204156261228786]	71/2000	1465

composition-20261013-185512-2bffd477

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [12, 13], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	-0.0005983137392528166	-0.0005957751251080846	[-0.000967770908744049 3, -0.00022507084570211577]	71/2000	1465
23	0.0004492915974461228	0.0004649281307566705 5	[5.583698743252717e-05, 0.0008733775102226093]	71/2000	1465
37	0.0001664124767705408 6	0.0001822159329953418 2	[-0.000260722413231894 8, 0.000609754279693661]	71/2000	1465

composition-20261013-185512-2bffd477

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [24, 25], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	8.042555401250508e-05	8.104170434346619e-05	[-7.955390526647082e-05, 0.000235949191167862 98]	71/2000	1465

composition-20261014-032120-486d799f

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [0], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	-4.60899341388781e-05	-4.0872599009493115e-0 5	[-0.000217734233338580 93, 0.00016858181205444938]	71/2000	1465
23	-3.271375526325137e-05	-3.199588131774163e-05	[-0.000174187935932211 99, 0.0001136384687997568]	71/2000	1465
37	-6.647979225569334e-05	-7.095429546745745e-05	[-0.000291885207220884 4, 9.73636943746164e-05]	71/2000	1465

composition-20261014-032120-486d799f

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [12], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	-0.0008545763004039009	-0.0008561104717573092	[-0.001150257172183655 8, -0.00056394938574460 14]	71/2000	1465
23	-9.848083648175 39e-05	-9.306537422449584e-05	[-0.000375805062010882 6, 0.00017279865033056432]	71/2000	1465

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
37	-7.626569138096745e-05	-7.009637664775361e-05	[-0.000360900941821140 16, 0.00023415362379323]	71/2000	1465

composition-20261014-032120-486d799f

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [25], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	0.0001729304994845724 8	0.0001719993236095868 5	[4.463885584017398e-05, 0.0002936832496752188]	71/2000	1465
23	-0.0007079791885041549	-0.000699445138327318	[-0.000889761639764939 4, -0.000521697610992854]	71/2000	1465
37	-7.243953146688717e-05	-6.619463365792254e-05	[-0.000212401684606083 8, 7.996392602977384e-05]	71/2000	1465

composition-20261014-032120-486d799f

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [0, 1], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	-0.0002764577707105269	-0.000305017713926657	[-0.000801674596175575 1, 0.00016777266329458 292]	71/2000	1465
23	0.0002748313234466563 5	0.0002560327026930727	[-0.000328662642760605 2, 0.0008294795855742868]	71/2000	1465
37	-0.0001416761346267915 2	-0.0001526383887316477 5	[-0.000643328925356078 7, 0.0003204156261228786]	71/2000	1465

composition-20261014-032120-486d799f

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [12, 13], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	-0.0005983137392528166	-0.0005957751251080846	[-0.000967770908744049 3, -0.00022507084570211 577]	71/2000	1465
23	0.0004492915974461228	0.0004649281307566705 5	[5.583698743252717e-05, 0.0008733775102226093]	71/2000	1465
37	0.0001664124767705408 6	0.0001822159329953418 2	[-0.000260722413231894 8, 0.000609754279693661]	71/2000	1465

composition-20261014-032120-486d799f

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [24, 25], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	8.042555401250508e-05	8.104170434346619e-05	[-7.955390526647082e-05 , 0.000235949191167862 98]	71/2000	1465
23	-0.0008111692747406037	-0.0008061970833859053	[-0.001029544344414319 8, -0.00058981680664782 52]	71/2000	1465
37	0.0001807916412591724 4	0.0001831437652353506 8	[-9.780179775247661e-06 , 0.0003823935562813924]	71/2000	1465

composition-20261014-032120-486d799f

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [0, 1, 2, 3], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	-0.0006571169305328121	-0.0006716296099761055	[-0.00164636577805492, 0.0001938152765730989 4]	71/2000	1465
23	7.183583249946473e-05	6.899960868660811e-05	[-0.000682006490905095 6, 0.0007963317650939533]	71/2000	1465
37	-0.0002972410869361785 3	-0.0003128500272375454 3	[-0.000933935280597861 5, 0.0002562151766575395]	71/2000	1465

composition-20261014-032120-486d799f

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [11, 12, 13, 14], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	0.003885308129297546	0.0039059187090884325	[0.0026608101493276353 , 0.0051295292904874204]	71/2000	1465
23	0.003589996487403734	0.003661064528190402	[0.002338269696115027, 0.004953754396441149]	71/2000	1465

composition-20261014-054423-dbd821ff

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [0], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	-4.60899341388781e-05	-4.0872599009493115e-0 5	[-0.000217734233338580 93, 0.0001685818120544 4938]	71/2000	1465
23	-3.271375526325137e-05	-3.199588131774163e-05	[-0.000174187935932211 99, 0.0001136384687997568]	71/2000	1465
37	-6.647979225569334e-05	-7.095429546745745e-05	[-0.000291885207220884 4, 9.73636943746164e-05]	71/2000	1465

composition-20261014-054423-dbd821ff

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [12], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	-0.0008545763004039009	-0.0008561104717573092	[-0.001150257172183655 8, -0.00056394938574460 14]	71/2000	1465
23	-9.84808364817539e-05	-9.306537422449584e-05	[-0.000375805062010882 6, 0.00017279865033056 432]	71/2000	1465
37	-7.626569138096745e-05	-7.009637664775361e-05	[-0.000360900941821140 16, 0.00023415362379323]	71/2000	1465

composition-20261014-054423-dbd821ff

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [25], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	0.0001729304994845724 8	0.0001719993236095868 5	[4.463885584017398e-05, 0.0002936832496752188]	71/2000	1465

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
23	-0.0007079791885041549	-0.000699445138327318	[-0.000889761639764939 4, -0.000521697610992854]	71/2000	1465
37	-7.243953146688717e-05	-6.619463365792254e-05	[-0.000212401684606083 8, 7.996392602977384e-05]	71/2000	1465

composition-20261014-054423-dbd821ff

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [0, 1], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	-0.0002764577707105269	-0.000305017713926657	[-0.000801674596175575 1, 0.00016777266329458292]	71/2000	1465
23	0.0002748313234466563 5	0.0002560327026930727	[-0.000328662642760605 2, 0.0008294795855742868]	71/2000	1465
37	-0.0001416761346267915 2	-0.0001526383887316477 5	[-0.000643328925356078 7, 0.0003204156261228786]	71/2000	1465

composition-20261014-054423-dbd821ff

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [12, 13], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	-0.0005983137392528166	-0.0005957751251080846	[-0.000967770908744049 3, -0.00022507084570211577]	71/2000	1465
23	0.0004492915974461228	0.0004649281307566705 5	[5.583698743252717e-05, 0.0008733775102226093]	71/2000	1465
37	0.0001664124767705408 6	0.0001822159329953418 2	[-0.000260722413231894 8, 0.000609754279693661]	71/2000	1465

composition-20261014-054423-dbd821ff

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [24, 25], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	8.042555401250508e-05	8.104170434346619e-05	[-7.955390526647082e-05 , 0.000235949191167862 98]	71/2000	1465
23	-0.0008111692747406037	-0.0008061970833859053	[-0.001029544344414319 8, -0.00058981680664782 52]	71/2000	1465
37	0.0001807916412591724 4	0.0001831437652353506 8	[-9.780179775247661e-06 , 0.0003823935562813924]	71/2000	1465

composition-20261014-054423-dbd821ff

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [0, 1, 2, 3], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	-0.0006571169305328121	-0.0006716296099761055	[-0.00164636577805492, 0.0001938152765730989 4]	71/2000	1465
23	7.183583249946473e-05	6.899960868660811e-05	[-0.000682006490905095 6, 0.0007963317650939533]	71/2000	1465

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
37	-0.0002972410869361785 3	-0.0003128500272375454 3	[-0.000933935280597861 5, 0.0002562151766575395]	71/2000	1465

composition-20261014-054423-dbd821ff

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [11, 12, 13, 14], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	0.003885308129297546	0.0039059187090884325	[0.0026608101493276353 , 0.0051295292904874204]	71/2000	1465
23	0.003589996487403734	0.003661064528190402	[0.002338269696115027, 0.004953754396441149]	71/2000	1465
37	0.0006958575765264769	0.000706815205475545	[-0.000485747489089789 36, 0.0019332813322892252]	71/2000	1465

composition-20261014-054423-dbd821ff

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Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	-0.0004223128795064941	-0.0004186258033244362 7	[-0.000905342181576156 4, 8.412072718029896e-05]	71/2000	1465
23	-0.00290206343154077	-0.0029018436309528315	[-0.003464295129582087, -0.0023463511594901273]	71/2000	1465
37	0.0003427221577525009	0.0003274063402903456	[-0.000174796254365639 24, 0.000831103017138969]	71/2000	1465

composition-20261014-054423-dbd821ff

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	-0.0031192516557841613	-0.003069222128221489	[-0.005598433325437209, -0.0005771062229211223]	71/2000	1465
23	-0.00867016563638367	-0.008560964606249673	[-0.01236161516313488, -0.004879390368432393]	71/2000	1465
37	-0.008888696495308254	-0.008889474302069626	[-0.011564302129920533, -0.006193025270745394]	71/2000	1465

Verified references

[1] [chen2021drone] Patrick Chen and Hsiang-Fu Yu and Inderjit S. Dhillon and Cho-Jui Hsieh. DRONE: Data-aware Low-rank Compression for Large NLP Models. Advances in Neural Information Processing Systems. 2021. https://proceedings.neurips.cc/paper/2021/hash/f56de5ef149cf0aedcc8f4797031e229-Abstract.html

[2] [hsu2022weighted] Yen-Chang Hsu and Ting Hua and Sung-En Chang and Qian Lou and Yilin Shen and Hongxia Jin. Language Model Compression with Weighted Low-Rank Factorization. International Conference on Learning Representations. 2022. Author spelling from conference PDF; arXiv:2207.00112 first submitted 2022. https://openreview.net/pdf?id=uPv9Y3gmAI5

[3] [yuan2023asvd] Zhihang Yuan and Yuzhang Shang and Yue Song and Dawei Yang and Qiang Wu and Yan Yan and Guangyu Sun. ASVD: Activation-aware Singular Value Decomposition for Compressing Large Language Models. 2023. First submitted 10 December 2023; verified revision v5 dated 28 August 2025. https://arxiv.org/abs/2312.05821

[4] [wang2024svdllm] Xin Wang and Yu Zheng and Zhongwei Wan and Mi Zhang. SVD-LLM: Truncation-aware Singular Value Decomposition for Large Language Model Compression. 2024. First submitted 12 March 2024; arXiv metadata identifies ICLR 2025; verified revision v5. https://arxiv.org/abs/2403.07378

[5] [wang2025svdllmv2] Xin Wang and Samiul Alam and Zhongwei Wan and Hui Shen and Mi Zhang. SVD-LLM V2: Optimizing Singular Value Truncation for Large Language Model Compression. Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers). 2025. https://aclanthology.org/2025.naacl-long.217/

[6] [liu2024eora] Shih-Yang Liu and Maksim Khadkevich and Nai Chit Fung and Charbel Sakr and Chao-Han Huck Yang and Chien-Yi Wang and Saurav Muralidharan and Hongxu Yin and Kwang-Ting Cheng and Jan Kautz and Yu-Chiang Frank Wang and Pavlo Molchanov and Min-Hung Chen. EoRA: Fine-tuning-free Compensation for Compressed LLM with Eigenspace Low-Rank Approximation. 2024. First submitted 28 October 2024; verified v6 dated 17 March 2026; arXiv metadata says ICLR 2026 workshops, not main conference. https://arxiv.org/abs/2410.21271

[7] [hu2026saessvd] Xing Hu and Dawei Yang and Yuan Cheng and Zhixuan Chen and Zukang Xu. SAES-SVD: Self-Adaptive Suppression of Accumulated and Local Errors for SVD-based LLM Compression. 2026. First submitted 3 February 2026; preprint, venue not verified. https://arxiv.org/abs/2602.03051

[8] [liu2026lacesvd] Zhuowen Liu and Longkun Hao and Shiyu Feng and Xiaowen Chang and Ruiqun Li and Changqun Li. LACE-SVD: Loss-Aware SVD with Cumulative Error Correction for LLM Compression. 2026. First submitted 3 July 2026; verified v2 dated 15 August 2026; preprint, venue not verified. https://arxiv.org/abs/2607.03057

[9] [abbasi2026zerosum] Ali Abbasi and Chayne Thrash and Haoran Qin and Shansita Sharma and Sepehr Seifi and Soheil Kolouri. Zero Sum SVD: Balancing Loss Sensitivity for Low Rank LLM Compression. 2026. First submitted 2 February 2026; verified v2 dated 28 September 2026; preprint, venue not verified. https://arxiv.org/abs/2602.02848

[10] [abbasi2026iosvd] Ali Abbasi and Chayne Thrash and Haoran Qin and Hamed Pirsiavash and Soheil Kolouri. IO-SVD: Input-Output Whitened SVD for Adaptive-Rank LLM Compression. 2026. First submitted 15 May 2026; preprint, venue not verified. https://arxiv.org/abs/2605.15626

[11] [arai2025qep] Yamato Arai and Yuma Ichikawa. Quantization Error Propagation: Revisiting Layer-Wise Post-Training Quantization. 2025. First submitted 13 April 2025; arXiv metadata says accepted at NeurIPS 2025; verified v3 dated 13 January 2026. https://arxiv.org/abs/2504.09629

[12] [frantar2022gptq] Elias Frantar and Saleh Ashkboos and Torsten Hoefler and Dan Alistarh. GPTQ: Accurate Post-Training Quantization for Generative Pre-trained Transformers. 2022. First submitted 31 October 2022; arXiv metadata identifies ICLR 2023; verified v2. https://arxiv.org/abs/2210.17323

[13] [lin2023awq] Ji Lin and Jiaming Tang and Haotian Tang and Shang Yang and Wei-Ming Chen and Wei-Chen Wang and Guangxuan Xiao and Xingyu Dang and Chuang Gan and Song Han. AWQ: Activation-aware Weight Quantization for LLM Compression and Acceleration. 2023. First submitted 1 June 2023; arXiv metadata identifies MLSys 2024; verified v6 dated 25 April 2026. https://arxiv.org/abs/2306.00978

[14] [frantar2023sparsegpt] Elias Frantar and Dan Alistarh. SparseGPT: Massive Language Models Can Be Accurately Pruned in One-Shot. 2023. First submitted 2 January 2023; verified v3; this entry cites the preprint. https://arxiv.org/abs/2301.00774

[15] [li2024svdquant] Muyang Li and Yujun Lin and Zhekai Zhang and Tianle Cai and Xiuyu Li and Junxian Guo and Enze Xie and Chenlin Meng and Jun-Yan Zhu and Song Han. SVDQuant: Absorbing Outliers by Low-Rank Components for 4-Bit Diffusion Models. 2024. First submitted 7 November 2024; arXiv metadata identifies ICLR 2025 Spotlight; verified v4; diffusion-model scope. https://arxiv.org/abs/2411.05007

[16] [kornblith2019cka] Simon Kornblith and Mohammad Norouzi and Honglak Lee and Geoffrey Hinton. Similarity of Neural Network Representations Revisited. Proceedings of the 36th International Conference on Machine Learning. 2019. https://proceedings.mlr.press/v97/kornblith19a.html

[17] [guo2017calibration] Chuan Guo and Geoff Pleiss and Yu Sun and Kilian Q. Weinberger. On Calibration of Modern Neural Networks. Proceedings of the 34th International Conference on Machine Learning. 2017. https://proceedings.mlr.press/v70/guo17a.html

[18] [biderman2024evaluation] Stella Biderman and Hailey Schoelkopf and Lintang Sutawika and Leo Gao and Jonathan Tow and Baber Abbasi and Alham Fikri Aji and Pawan Sasanka Ammanamanchi and Sidney Black and Jordan Clive and Anthony DiPofi and Julen Etxaniz and Benjamin Fattori and Jessica Zosa Forde and Charles Foster and Jeffrey Hsu and Mimansa Jaiswal and Wilson Y. Lee and Haonan Li and Charles Lovering and Niklas Muennighoff and Ellie Pavlick and Jason Phang and Aviya Skowron and Samson Tan and Xiangru Tang and Kevin A. Wang and Genta Indra Winata and Fran\c cois Yvon and Andy Zou. Lessons from the Trenches on Reproducible Evaluation of Language Models. 2024. First submitted 23 May 2024; author list verified against v3 dated 31 May 2026; this entry cites the preprint. https://arxiv.org/abs/2405.14782

[19] [merity2016wikitext] Stephen Merity and Caiming Xiong and James Bradbury and Richard Socher. Pointer Sentinel Mixture Models. 2016. First submitted 26 September 2016; this entry cites the WikiText-introducing preprint. https://arxiv.org/abs/1609.07843

[20] [raffel2020t5] Colin Raffel and Noam Shazeer and Adam Roberts and Katherine Lee and Sharan Narang and Michael Matena and Yanqi Zhou and Wei Li and Peter J. Liu. Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer. Journal of Machine Learning Research. 2020. Journal publication 2020; original arXiv:1910.10683 first submitted 23 October 2019. https://jmlr.org/papers/v21/20-074.html

[21] [zellers2019hellaswag] Rowan Zellers and Ari Holtzman and Yonatan Bisk and Ali Farhadi and Yejin Choi. HellaSwag: Can a Machine Really Finish Your Sentence?. Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics. 2019. https://aclanthology.org/P19-1472/

[22] [clark2018arc] Peter Clark and Isaac Cowhey and Oren Etzioni and Tushar Khot and Ashish Sabharwal and Carissa Schoenick and Oyvind Tafjord. Think you have Solved Question Answering? Try ARC, the AI2 Reasoning Challenge. 2018. First submitted 14 March 2018; this entry cites the preprint. https://arxiv.org/abs/1803.05457

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